Summary: Variables That Affect Demand

Variable	A change in this variable…
Price	…causes a movement along the **D** curve
No. of buyers	…shifts the **D** curve
Income	…shifts the **D** curve
Price of related goods	…shifts the **D** curve
Tastes	…shifts the **D** curve

Summary: Variables That Affect Supply

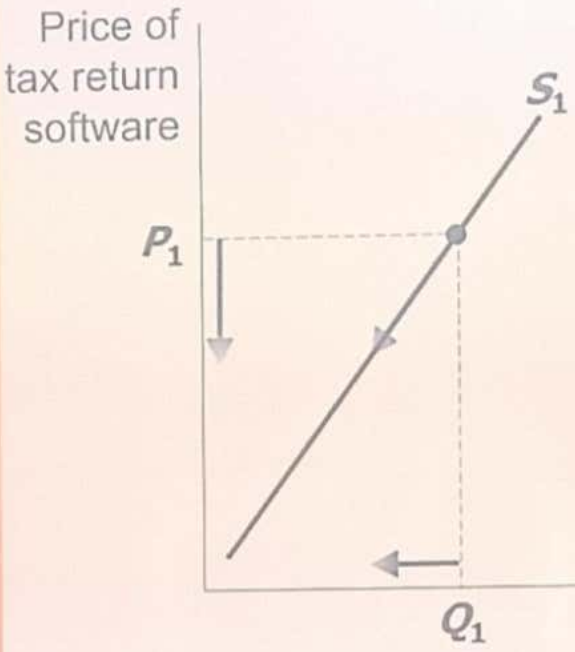
Variable	A change in this variable…
Price	…causes a movement along the **S** curve
Input prices	…shifts the **S** curve
Technology	…shifts the **S** curve
No. of sellers	…shifts the **S** curve
Expectations	…shifts the **S** curve

Summary: Variables That Affect Supply

Variable	A change in this variable…
Price	…causes a movement along the **S** curve
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Expectations	…shifts the **S** curve

ACTIVE LEARNING 2:

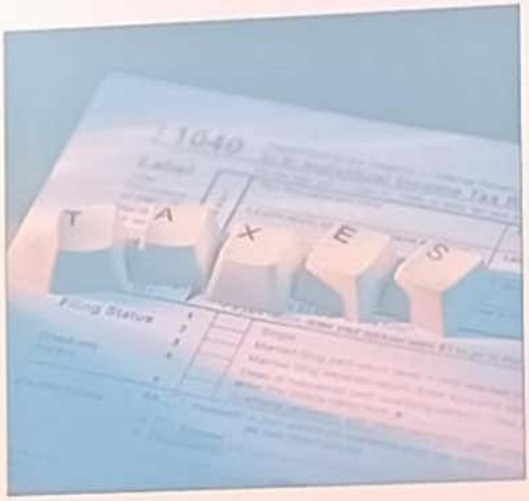
A. fall in price of tax return software



The **S** curve does not shift.

Move down along the curve to a lower ***P*** and lower ***Q***.

ACTIVE LEARNING 2:
Supply curve



Draw a supply curve for tax return preparation software. What happens to it in each of the following scenarios?

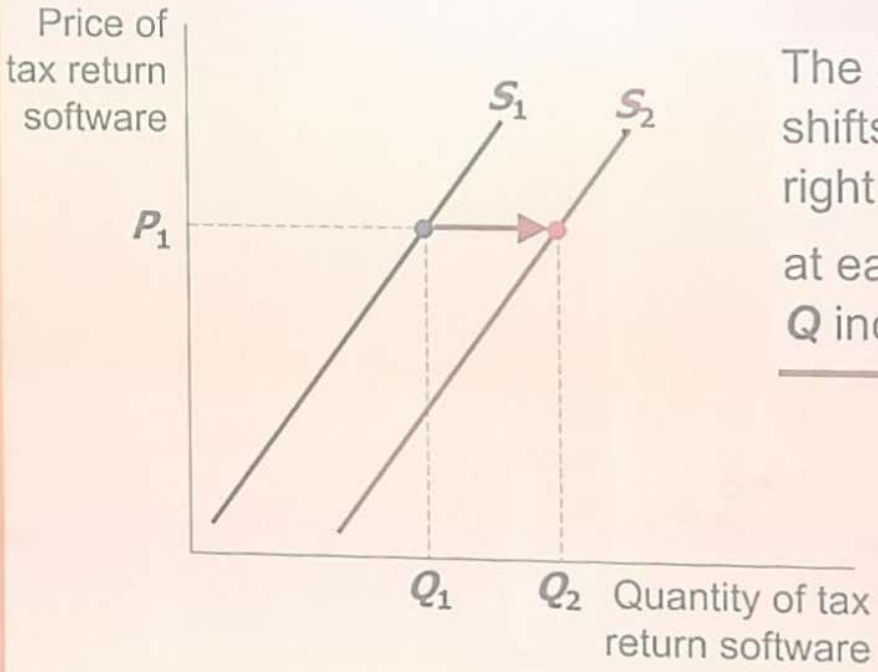
A. Retailers cut the price of the software.

B. A technological advance allows the software to be produced at lower cost.

C. Professional tax return preparers raise the price of the services they provide.

ACTIVE LEARNING 2:

B. fall in cost of producing the software

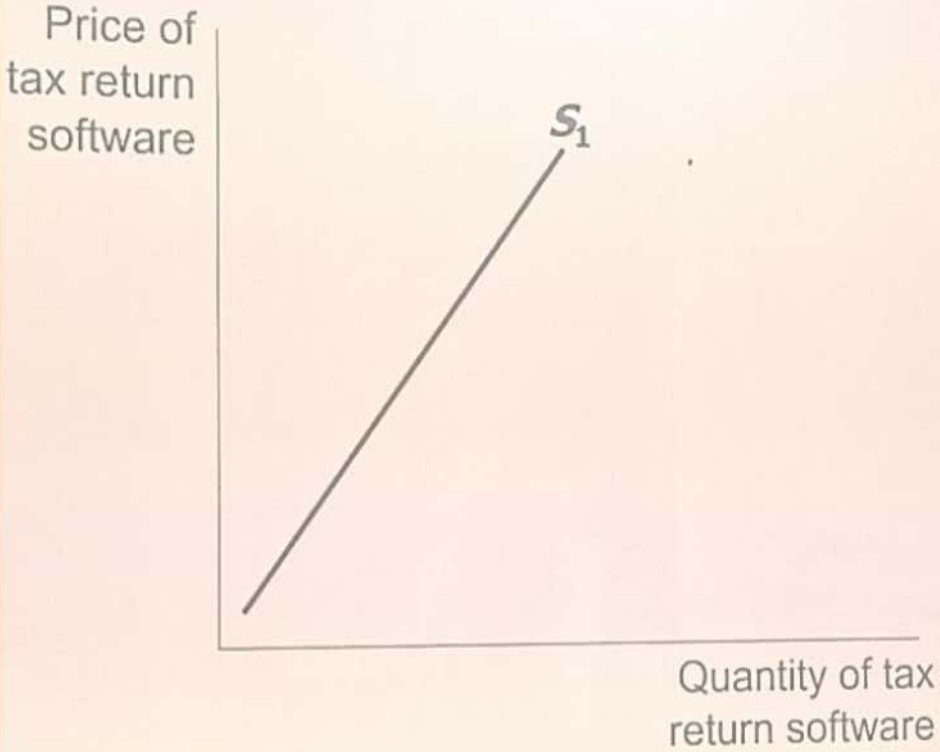


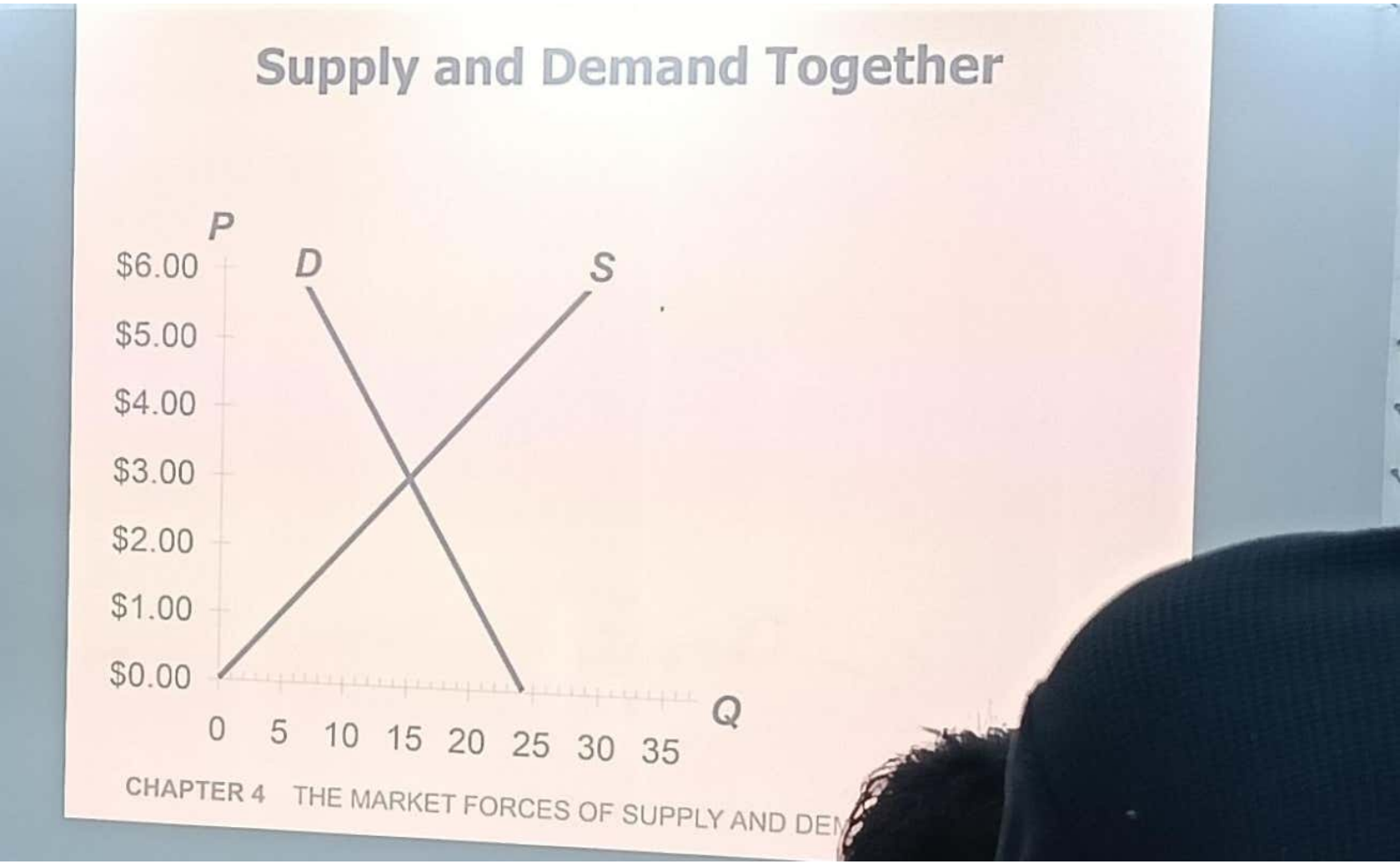
The **S** curve shifts to the right:

at each price, *Q* increases.

ACTIVE LEARNING 2:

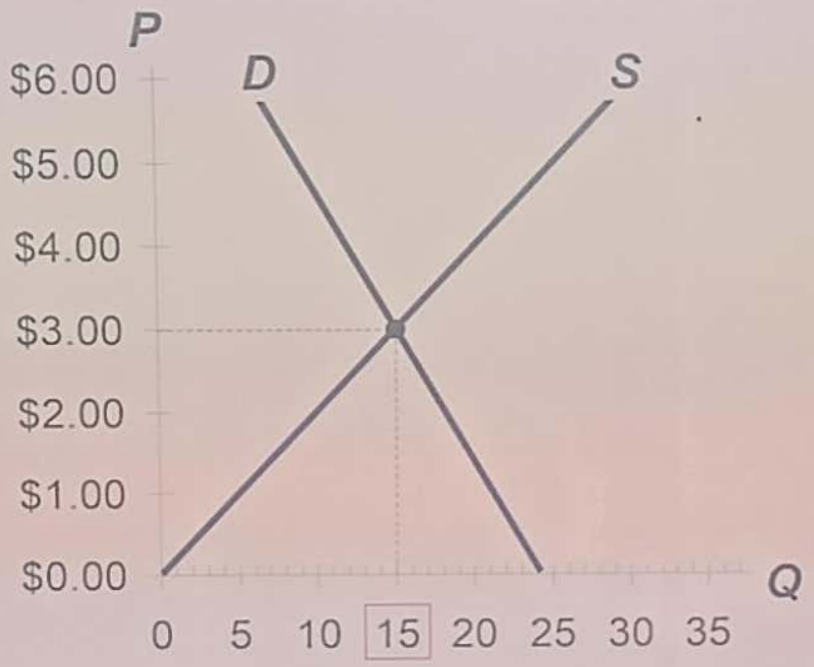
C. professional preparers raise their price




Supply and Demand Together
P
$6.00
$5.00
$4.00
$3.00
$2.00
$1.00
$0.00
D
S
Q
0 5 10 15 20 25 30 35
CHAPTER 4 THE MARKET FORCES OF SUPPLY AND DE

Equilibrium quantity:

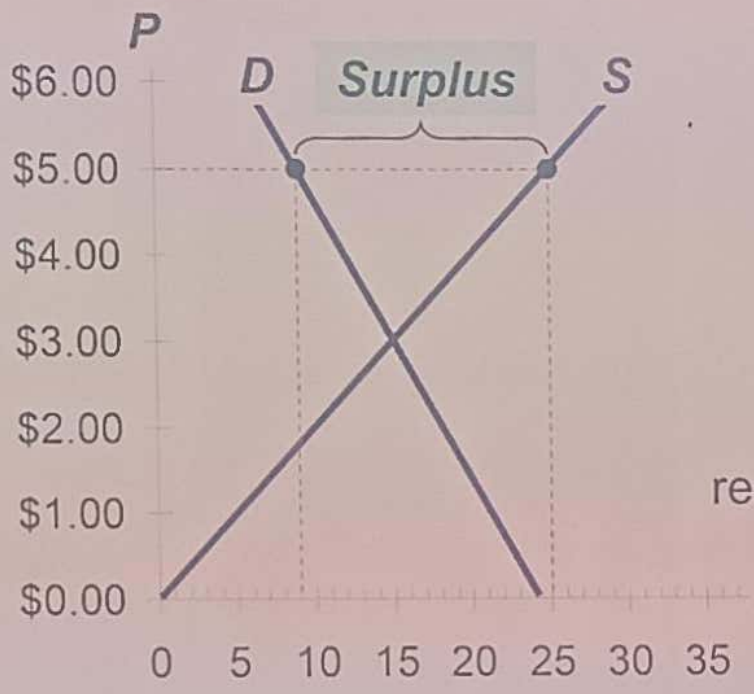
The quantity supplied and quantity demanded at the equilibrium price



P	Q^D	Q^S
$0	24	0
1	21	5
2	18	10
3	15	15
4	12	20
5	9	
6	6	

Surplus:

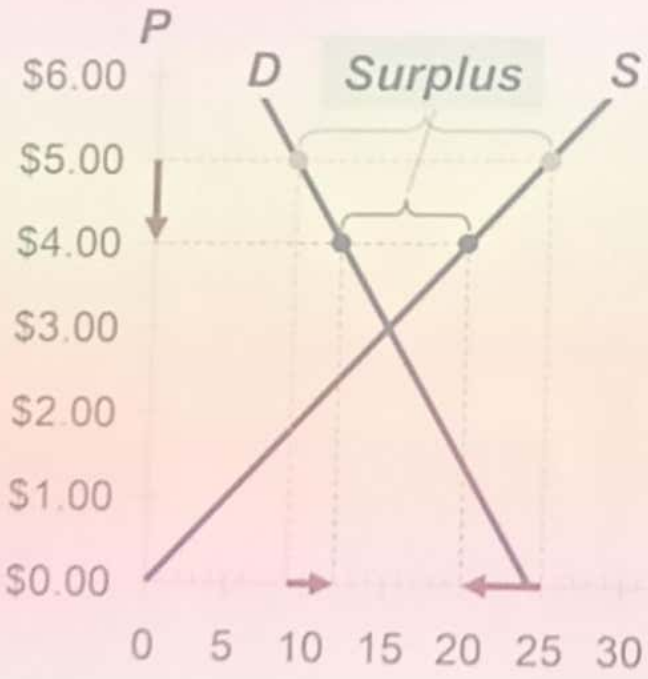
when quantity supplied is greater than quantity demanded



Example:
If P = \$5,
then
Q^D = 9 lattes
and
Q^S = 25 lattes
resulting in a surpl
of 16 la

Surplus:

when quantity supplied is greater than quantity demanded



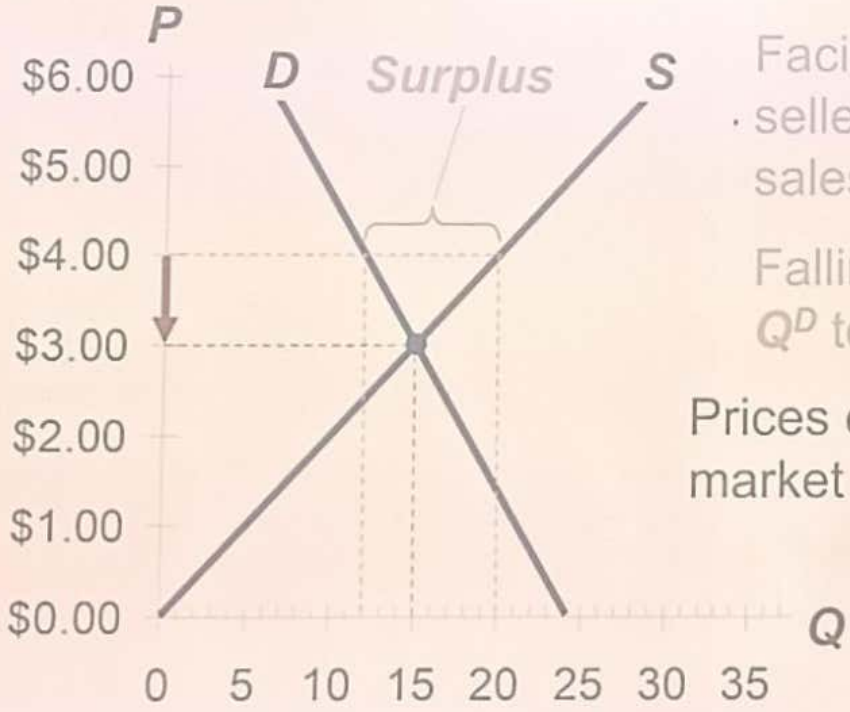
Facing a surplus, sellers try to increase sales by cutting the price.

This causes Q^D to rise and Q^S to fall…

…which reduces the surplus.

Surplus:

when quantity supplied is greater than quantity demanded

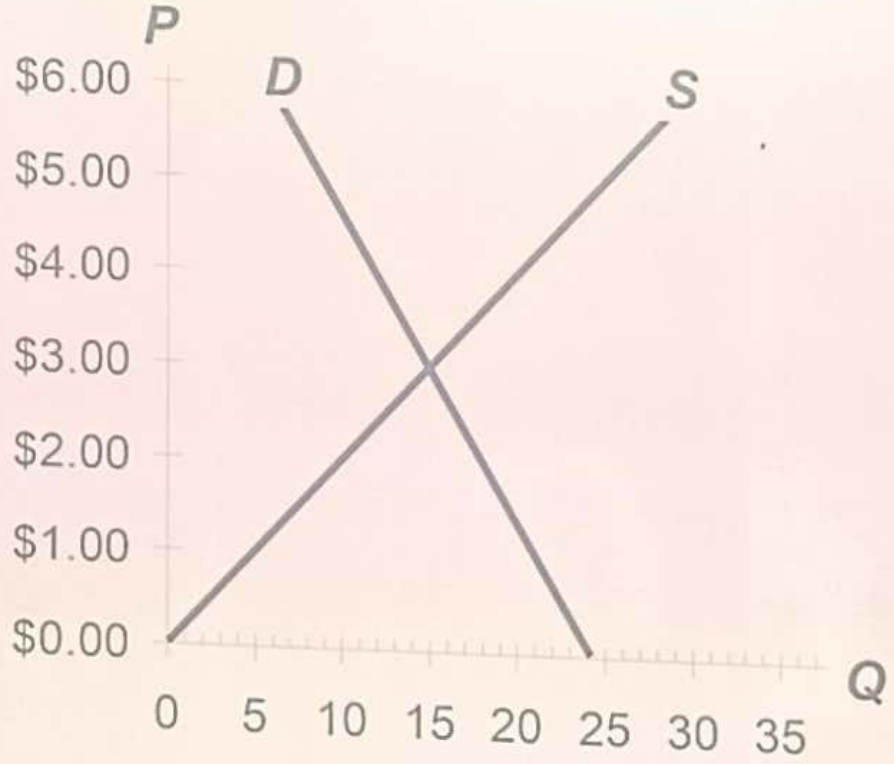


- Facing a surplus, sellers try to increase sales by cutting the price.
- Falling prices cause Q^D to rise and Q^S to fall.

Prices continue to fall until market reaches equilibrium.

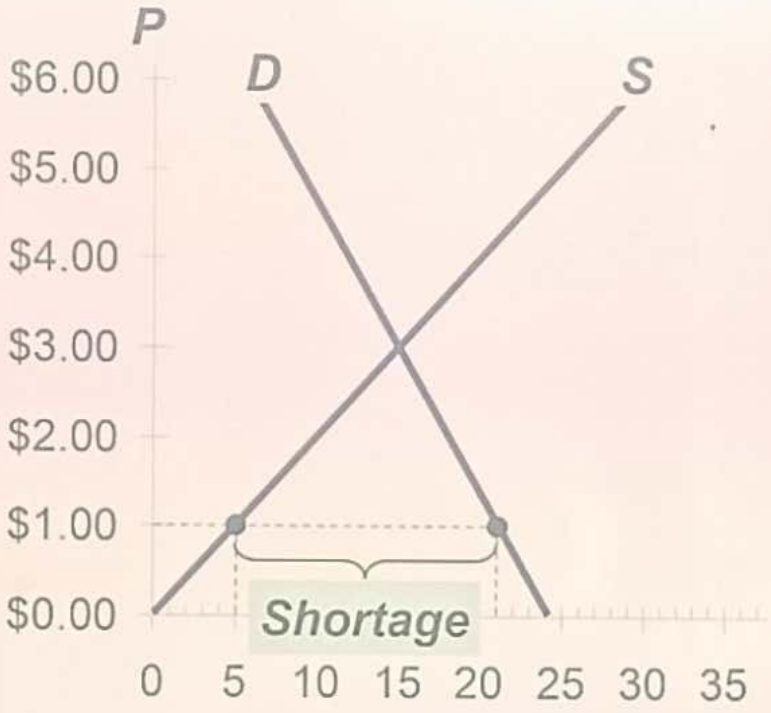
Shortage:

when quantity demanded is greater than quantity supplied



Shortage:

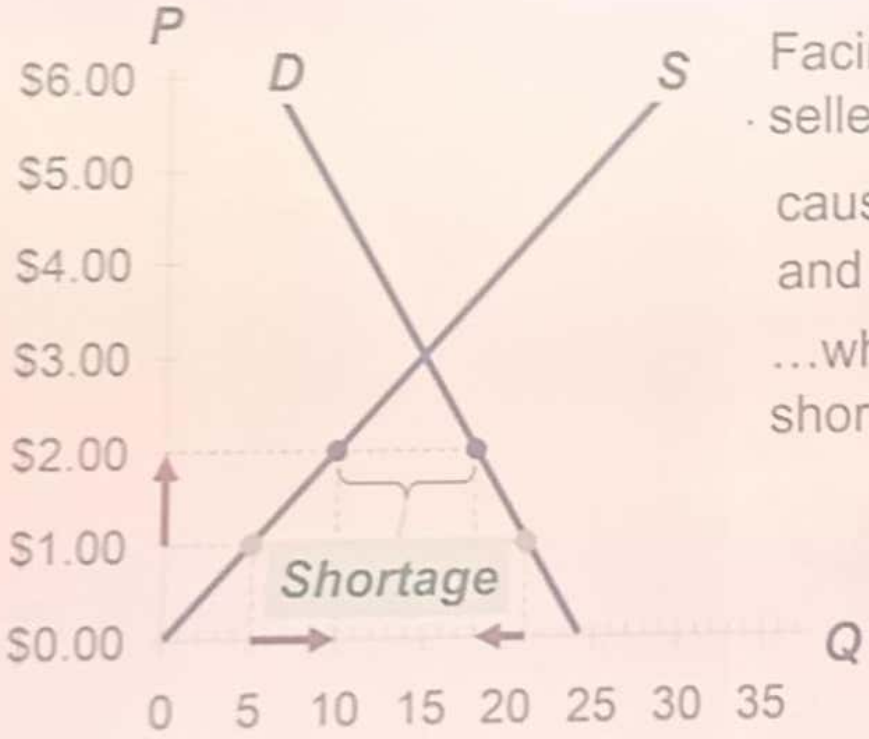
when quantity demanded is greater than quantity supplied



Example:
If P = \$1,
then
Q^D = 21 lattes
and
Q^S = 5 lattes
resulting in a shortage of 16 lattes

Shortage:

when quantity demanded is greater than quantity supplied



Facing a shortage, sellers raise the price,

causing Q^D to fall and Q^S to rise,

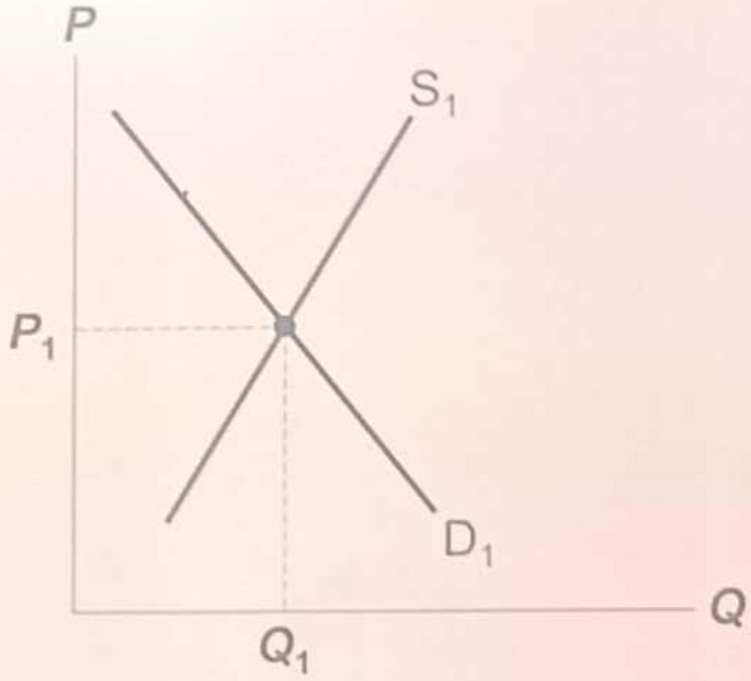
…which reduces the shortage.

Three Steps to Analyzing Changes in Eq'm

To determine the effects of any event,

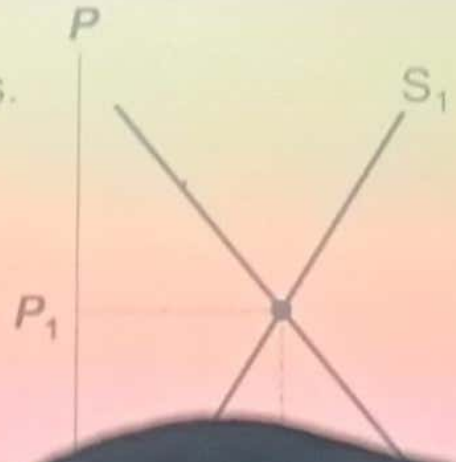
1. Decide whether event shifts **S** curve, **D** curve, or both.
2. Decide in which direction curve shifts.
3. Use supply-demand diagram to see how the shift changes eq'm ***P*** and ***Q***.

EXAMPLE 1: A Change in Demand



EXAMPLE 1: A Change in Demand

EVENT TO BE ANALYZED:
Increase in price of gas.



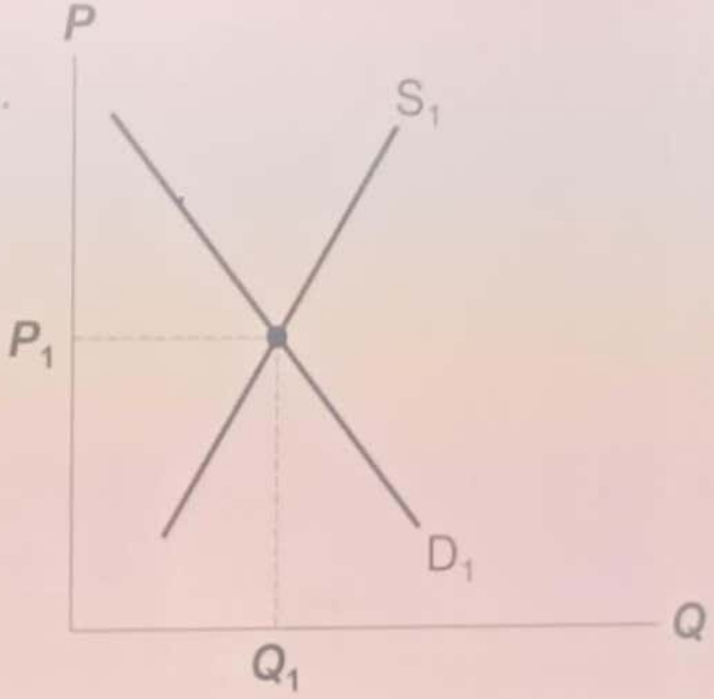
EXAMPLE 1: A Change in Demand

EVENT TO BE ANALYZED:
Increase in price of gas.

STEP 1:

D curve shifts because price of gas affects demand for hybrids.

S curve does not shift, because price of gas does not affect cost of producing hybrids.



EXAMPLE 1: A Change in Demand

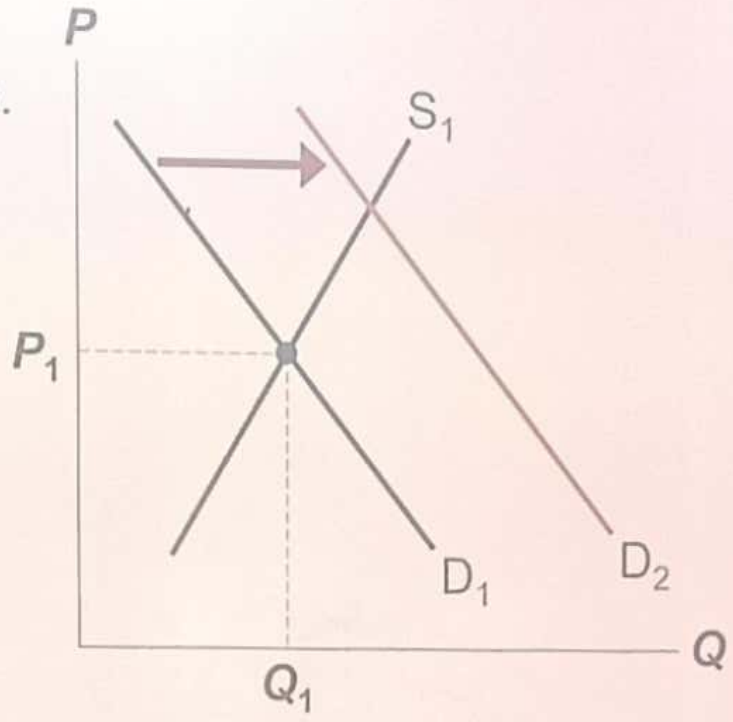
EVENT TO BE ANALYZED:
Increase in price of gas.

STEP 1:

D curve shifts

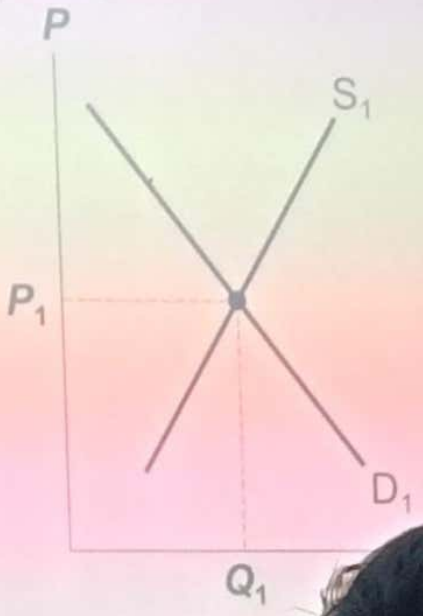
STEP 2:

D shifts right because high gas price makes hybrids more attractive relative to other cars.



EXAMPLE 2: A Change in Supply

EVENT: New technology reduces cost of producing hybrid cars.



EXAMPLE 2: A Change in Supply

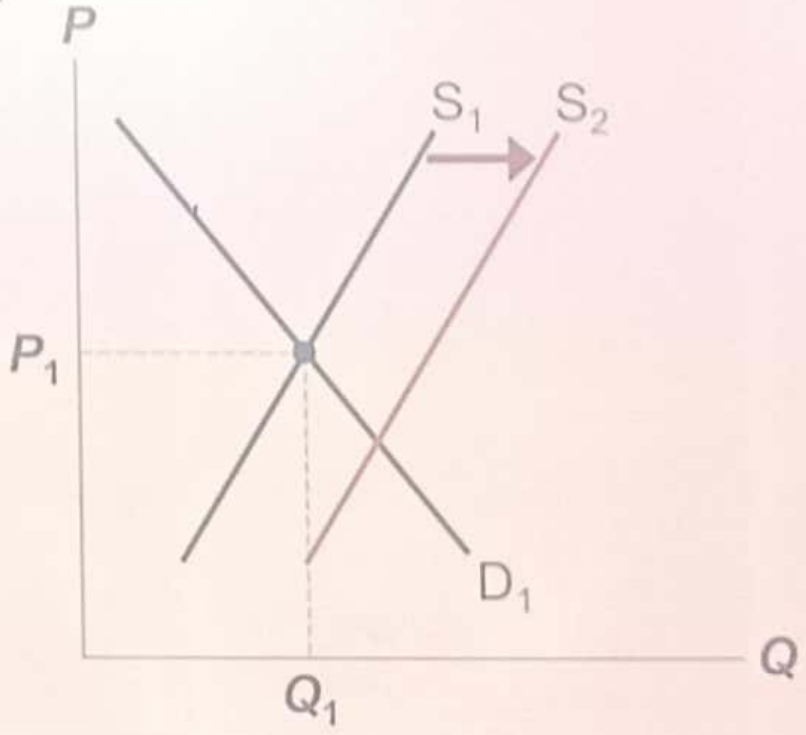
EVENT: New technology reduces cost of producing hybrid cars.

STEP 1:

S curve shifts

STEP 2:

S shifts <u>right</u> because event reduces cost, makes production more profitable at any given price.



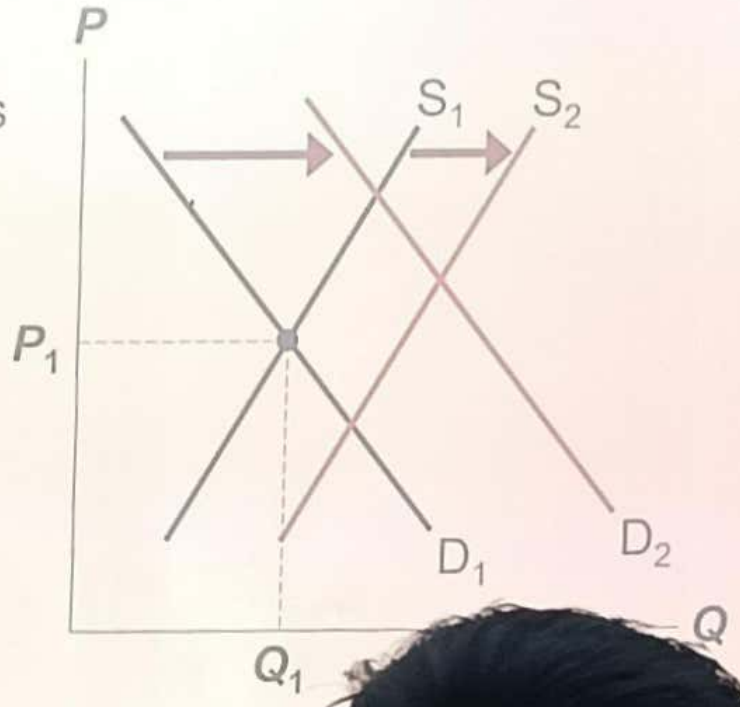
EXAMPLE 3: A Change in Both Supply and Demand

EVENTS:
price of gas rises AND
new technology reduces
production costs

STEP 1:
Both curves shift.

STEP 2:
Both shift to the right.

STEP 3:
Q rises, but effect
on *P* is ambiguous:

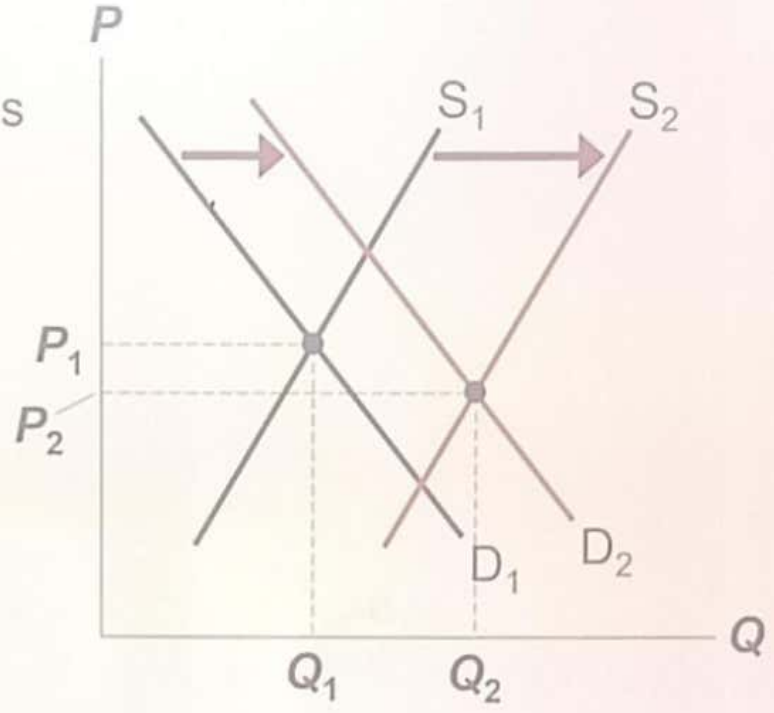


EXAMPLE 3: A Change in Both Supply and Demand

EVENTS:
price of gas rises AND
new technology reduces
production costs

STEP 3, cont.

But if supply increases more than demand, *P* falls.



ACTIVE LEARNING 3:
Changes in supply and demand

Use the three-step method to analyze the effects of each event on the equilibrium price and quantity of music downloads.

Event A: A fall in the price of compact discs

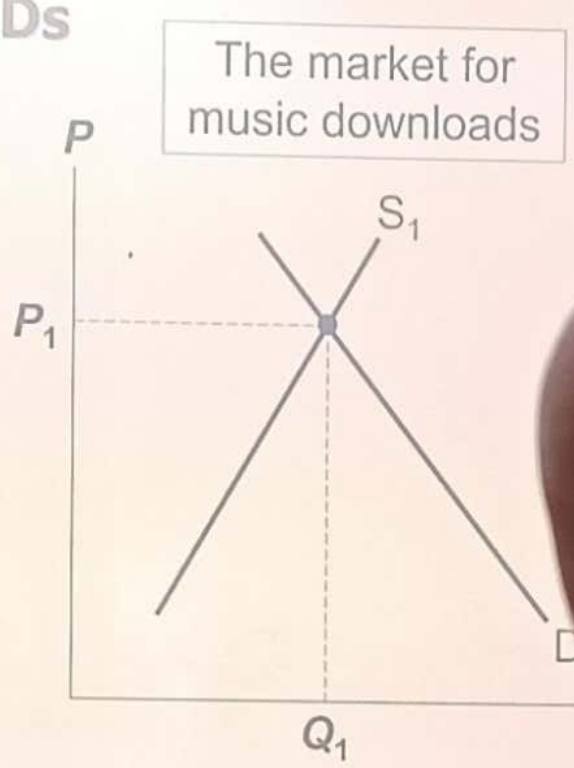
Event B: Sellers of music downloads negotiate a reduction in the royalties they must pay for each song they sell.

Event C: Events A and B both occur.

ACTIVE LEARNING 3:

A. fall in price of CDs

STEPS



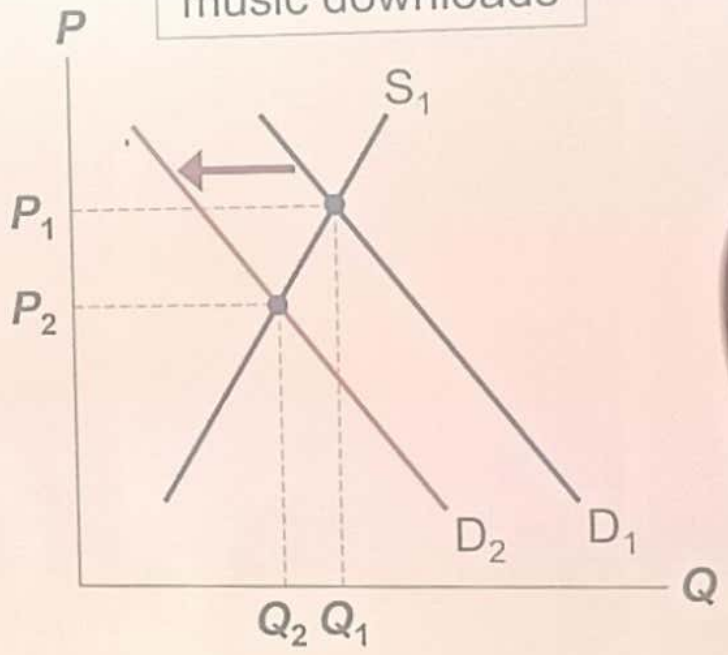
ACTIVE LEARNING 3:

A. fall in price of CDs

STEPS

1. **D** curve shifts
2. **D** shifts left
3. ***P*** and ***Q*** both fall.

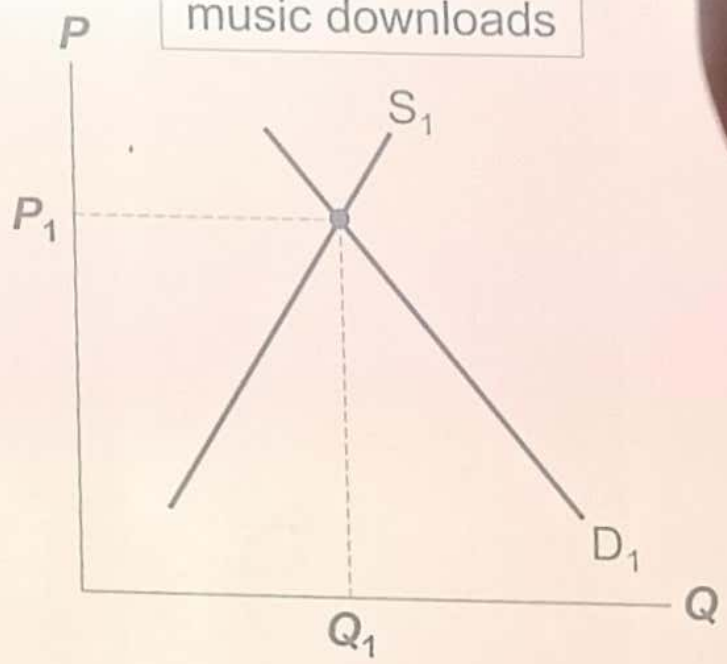
The market for music downloads



ACTIVE LEARNING 3:

B. fall in cost of royalties

STEPS

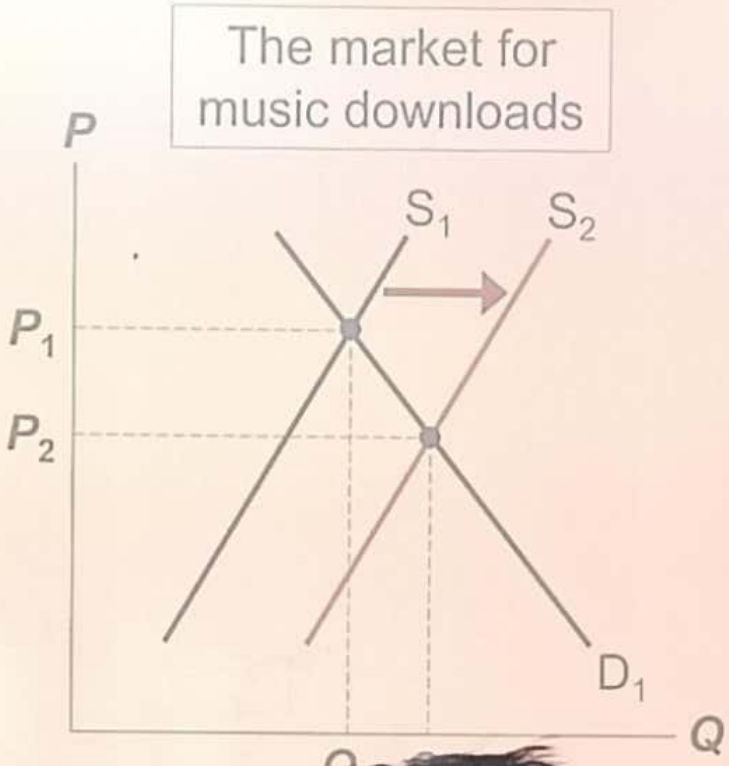


ACTIVE LEARNING 3:

B. fall in cost of royalties

STEPS

1. **S** curve shifts
2. **S** shifts right
3. ***P*** falls, ***Q*** rises.

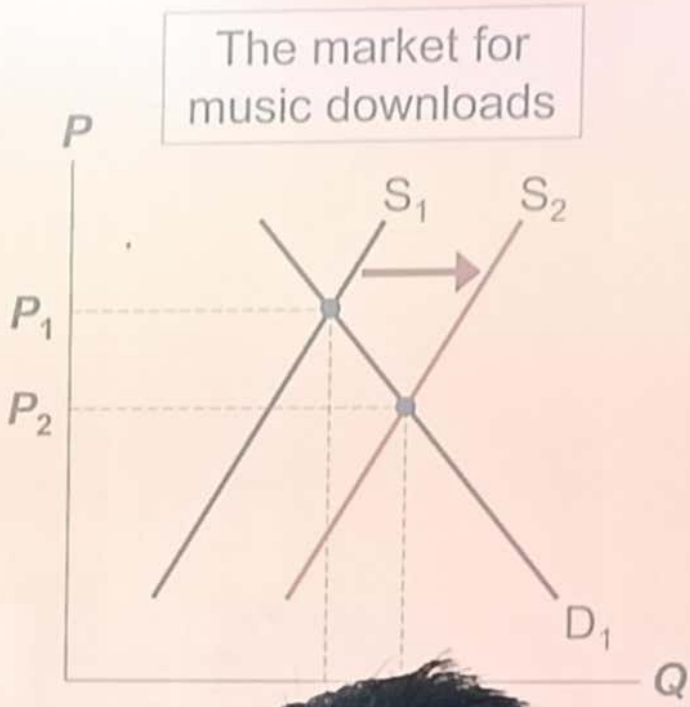


ACTIVE LEARNING 3:

B. fall in cost of royalties

STEPS

1. **S** curve shifts
2. **S** shifts right
3. ***P*** falls, ***Q*** rises.

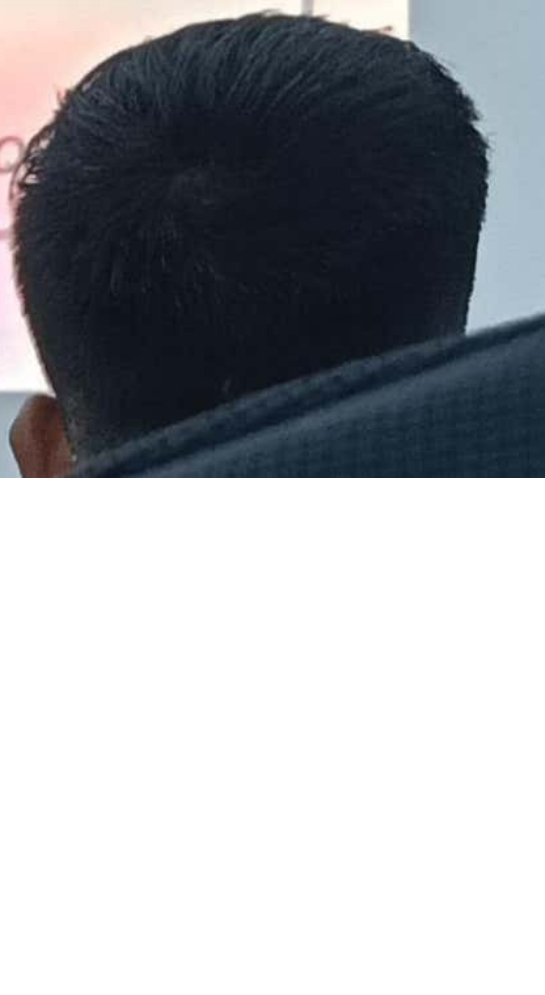


ACTIVE LEARNING 3:

C. fall in price of CDs AND fall in cost of royalties

STEPS

1. Both curves shift (see parts A & B).
2. **D** shifts left, **S** shifts right.
3. ***P*** unambiguously falls.
   Effect on ***Q*** is ambiguous:
   The fall in demand reduces ***Q***.
   the increase in supply increases ***Q***



A scenario...

You design websites for local businesses. You charge $200 per website, and currently sell 12 websites per month.

Your costs are rising (including the opp. cost of your time), so you're thinking of raising the price to $250.

The law of demand says that you won't sell as many websites if you raise your price. How many fewer websites? How much will your revenue fall, or might it increase?

Elasticity

- Basic idea: Elasticity measures how much one variable responds to changes in another variable.
  - One type of elasticity measures how much demand for your websites will fall if you raise your price.

Elasticity

- Basic idea: Elasticity measures how much one variable responds to changes in another variable.
  - One type of elasticity measures how much demand for your websites will fall if you raise your price.
- Definition:
  Elasticity is a numerical measure of the responsiveness of Q^d or Q^s to one of its determinants.

Price Elasticity of Demand

$$\text{Price elasticity of demand} = \frac{\text{Percentage change in } Q^d}{\text{Percentage change in } P}$$

- **Price elasticity of demand** measures how much Q^d responds to a change in P.

Price Elasticity of Demand

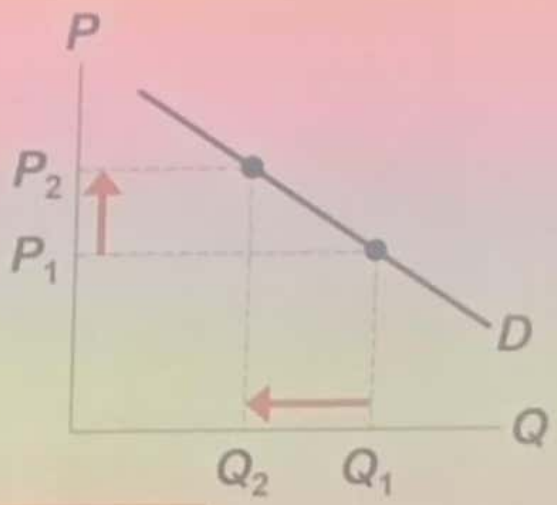
$$\text{Price elasticity of demand} = \frac{\text{Percentage change in } Q^d}{\text{Percentage change in } P}$$

Example:

Price elasticity of demand equals

$$\frac{15\%}{10\%} = 1.5$$

P rises by 10%



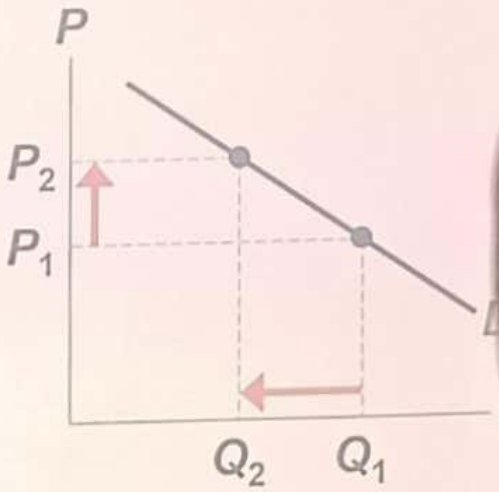
Q falls by 15%

Price Elasticity of Demand

$$\text{Price elasticity of demand} = \frac{\text{Percentage change in } Q^d}{\text{Percentage change in } P}$$

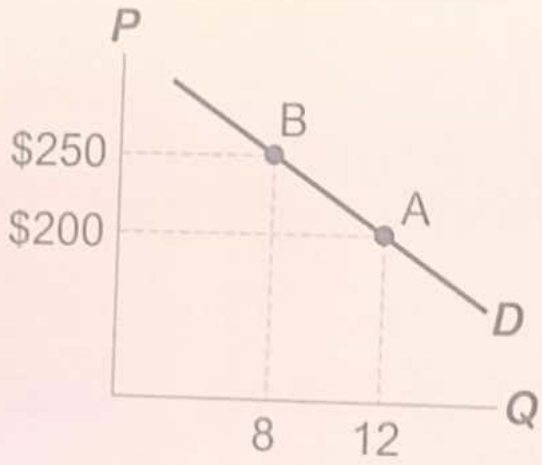
Along a ***D*** curve, ***P*** and ***Q*** move in opposite directions, which would make price elasticity negative.

We will drop the minus sign and report all price elasticities as positive numbers.



Calculating Percentage Changes

Demand for your websites



Standard method of computing the percentage (%) change:

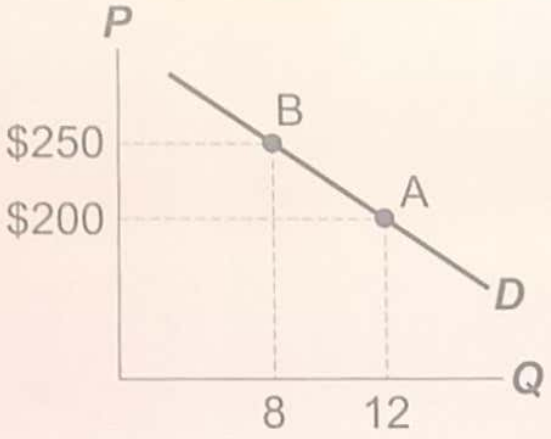
$$\frac{\text{end value} - \text{start value}}{\text{start value}} \times 100\%$$

Going from A to B, the % change in ***P*** equals

($250–$200)/$200 = 25%

Calculating Percentage Changes

Demand for your websites



Problem:

The standard method gives different answers depending on where you start.

From A to B,
P rises 25%, ***Q*** falls 33%
elasticity = 33/25 = 1.33

From B to A,
P falls 20%, ***Q*** rises 50%,
elasticity = 50/20 = 2.50

Calculating Percentage Changes

- Using the midpoint method, the % change in ***P*** equals

$$\frac{\$250 - \$200}{\$225} \times 100\% = \boxed{22.2\%}$$

- The % change in ***Q*** equals

$$\frac{12 - 8}{10} \times 100\% = \boxed{40.0\%}$$

Calculating Percentage Changes

- Using the midpoint method, the % change in ***P*** equals

$$\frac{\$250 - \$200}{\$225} \times 100\% = \boxed{22.2\%}$$

- The % change in ***Q*** equals

$$\frac{12 - 8}{10} \times 100\% = \boxed{40.0\%}$$

- The price elasticity of demand equals

$$40/22.2 = \boxed{1.8}$$

What determines price elasticity?

- Use midpoint method to calculate % change in Q^d

  (5000 – 3000)/4000 = 50%

- % change in P

  ($80 – $70)/$80 = 25%

- The price elasticity of demand equals

$$\frac{50\%}{25\%} = 2.0$$

What determines price elasticity?

To learn the determinants of price elasticity, we look at a series of examples.
Each compares two common goods.

In each example:

What determines price elasticity?

To learn the determinants of price elasticity, we look at a series of examples.
Each compares two common goods.

In each example:

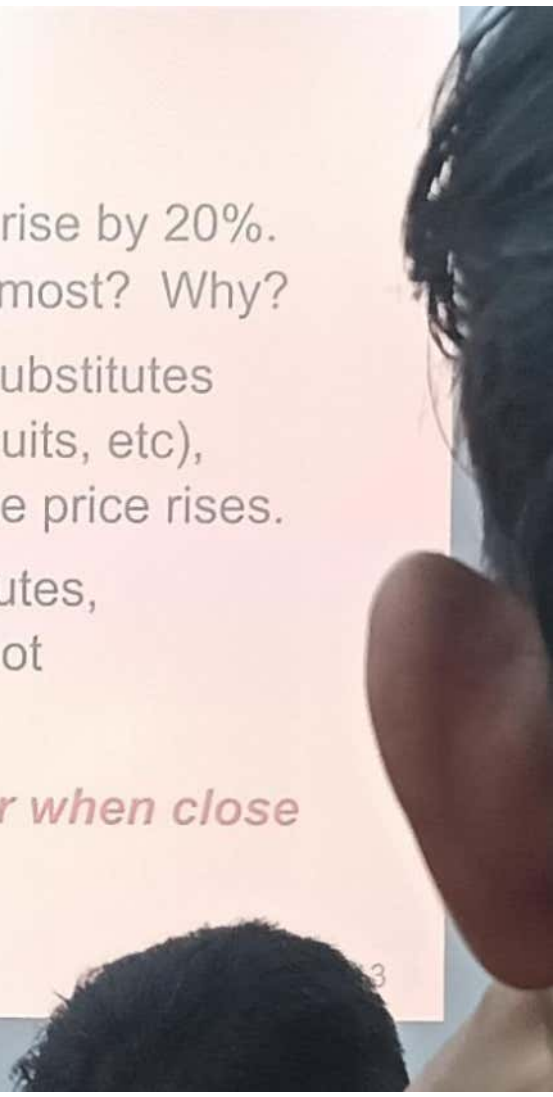
- Suppose the prices of both goods rise by 20%.
- The good for which Q^d falls the most (in percent) has the highest price elasticity of demand. Which good is it? Why?
- What lesson does the example teach us about the determinants of the price elasticity of demand?

EXAMPLE 1:
Potato chips vs. Sunscreen

- The prices of both of these goods rise by 20%. For which good does Q^d drop the most? Why?

EXAMPLE 1:
Potato chips vs. Sunscreen

- The prices of both of these goods rise by 20%. For which good does Q^d drop the most? Why?
  - Potato Chips has lots of close substitutes (*e.g.*, other chips- Kurkure, Biscuits, etc), so buyers can easily switch if the price rises.
  - Sunscreen has no close substitutes, so consumers would probably not buy much less if its price rises.
- Lesson: ***Price elasticity is higher when close substitutes are available.***



EXAMPLE 2:
"Tata Nexon" vs. "Cars"

- The prices of both goods rise by 20%.
  For which good does Q^d drop the most? Why?
  - For a narrowly defined good such as a car brand/model, there are many substitutes .
  - There are fewer substitutes available for broadly defined goods such as cars.
  - Lesson: *Price elasticity is higher for narrowly defined goods than broadly defined ones.*

EXAMPLE 3:
Insulin vs. Cruises

- The prices of both of these goods rise by 20%. For which good does Q^d drop the most? Why?
  - To millions of diabetics, insulin is a necessity. A rise in its price would cause little or no decrease in demand.

EXAMPLE 4:
Gasoline in the Short Run vs. Gasoline in the Long Run

- The price of gasoline rises 20%. Does Q^d drop more in the short run or the long run? Why?
  - There's not much people can do in the short run, other than ride the bus or carpool.
  - In the long run, people can buy smaller cars or live closer to where they work.

The Determinants of Price Elasticity: A Summary

The price elasticity of demand depends on:

- the extent to which close substitutes are available
- whether the good is a necessity or a luxury
- how broadly or narrowly the good is defined
- the time horizon: elasticity is higher in the long run than the short run.